

KAVYASRI KAMISSETTY

Data Analyst

USA | kavk38@gmail.com | (281) 789-8407 | [LinkedIn](#)

SUMMARY

- **Data Analyst with 4+ years of experience delivering advanced healthcare analytics across claims and EHR datasets, supporting risk prediction, cost optimization, and care-quality initiatives.**
- Strong expertise in data cleaning, validation, normalization, and descriptive statistics (mean, median, SD, percentiles) to ensure analytically sound outcomes. Adept at creating data visualizations using **Tableau and Power BI**.
- Proven ability to build hypothesis-driven analyses, correlation and regression models, and machine-learning solutions for **readmission risk, fraud detection, and chronic condition management** within HIPAA-compliant environments.
- Experienced in working with **ICD-9, ICD-10, CPT, HCPCS**, HL7-enabled clinical data, and CMS-aligned reporting requirements. Proven expertise in utilizing **Python, SQL, and R** for data manipulation and analysis.
- Advanced SQL practitioner (CTEs, subqueries, window functions) with hands-on experience integrating **claims vs. EHR data** from **epic**, translating complex healthcare data into actionable insights for clinical, financial, and operational stakeholders.

TECHNICAL SKILLS

Programming & Querying: Python (Pandas, NumPy, Scikit-learn), SQL (CTEs, Subqueries, Window Functions), R (Tidyverse)

Data Engineering & ETL: Apache Spark, PySpark, Apache Airflow, Databricks, KNIME, SSIS, SSRS, Power Query

Databases & Warehouse: Snowflake, Amazon Redshift, AWS EC2, Azure Synapse Analytics, Google BigQuery

Data Analysis & Statistics: Data Cleaning & Validation, Data Transformation, Normalization, Descriptive Statistics (Mean, Median, Standard Deviation, Percentiles), Hypothesis Testing, Correlation Analysis, Regression Analysis, Exploratory Data Analysis (EDA),

Databases: Oracle, MySQL, PostgreSQL, SQL Server, NoSQL (MongoDB)

Business Intelligence & Visualization: Tableau, Power BI, Qlik Sense, Looker, Microsoft Excel (DAX, Pivot, Power Query)

Modeling & Machine Learning: SAS, Predictive Modeling, A/B Testing, Time-Series Forecasting, Clustering and Segmentation

Data Governance & Compliance: Data Governance, Metadata Management(MDM), Data Quality Controls, HIPAA-Compliant Data Handling, CMS Reporting Readiness

Advanced Analytics & ML Tools: Risk Prediction Models, Readmission Models, Fraud Detection Models, Cost Optimization Models, Predictive Analytics, Feature Engineering, Model Validation

Big Data & Distributed Computing: Hadoop Ecosystem, Apache Spark, Spark SQL, Delta Lake

Healthcare & Domain Expertise: Claims and EHR/EMR Analytics, ICD-10, CPT, HCPCS, HL7/FHIR Messaging, Cerner, Epic Clarity, CMS Data Models, HIPAA Compliance, Clinical Quality and Cost Analytics

Version Control & Deployment: Git, GitHub, CI/CD Pipelines, API Data Integration, JSON and XML Processing

EXPERIENCE

Data Analyst | Cigna Healthcare - USA

Jun 2024 – Present

- Performed rigorous data cleaning, validation, transformation, and normalization on large-scale claims and **EHR** datasets, ensuring **HIPAA**-compliant, analytics-ready data for downstream modeling workflows.
- Conducted descriptive statistical analysis (mean, median, standard deviation, percentiles) to profile utilization, cost, and risk patterns across diverse member cohorts. Leveraged **AWS EC2** to deploy analytical models and run data processing tasks in a scalable environment, ensuring efficient handling of 90% large EHR datasets from Epic for real-time analysis.
- Developed risk prediction and readmission models using **Python**, leveraging **ICD-10** diagnosis codes, clinical indicators, and utilization signals to identify high-risk populations.
- Applied hypothesis testing, correlation analysis, and basic regression techniques to evaluate drivers of hospital readmissions, care gaps, and cost variability, **ANOVA** to compare the means of different patient groups, identifying significant differences in treatment outcomes and guiding targeted healthcare strategies
- Delivered **ad-hoc** analysis on healthcare and operational datasets, surfacing trends and actionable insights that informed strategic decisions on patient care optimization and resource allocation.
- Managed and automated data workflows with **Apache Airflow**, integrating Epic EHR and other healthcare data sources to ensure 60% faster, timely updates and seamless reporting.
- Applied **NLP** techniques (tokenization, sentiment analysis, entity recognition) on physician notes and patient feedback to uncover patterns in unstructured healthcare data, enhancing decision-making and patient care strategies.
- Designed and maintained interactive dashboards in Power BI & Tableau to track **KPIs** on patient care, hospital efficiency, and cost-effectiveness, enabling stakeholders to monitor trends and optimize resource allocation.
- Worked with **HL7**-enabled clinical feeds and Cerner-sourced datasets, aligning clinical attributes to standardized healthcare and CMS-aligned data models.
- Designed and executed **A/B testing** experiments to evaluate the effectiveness of healthcare interventions, providing

evidence-based recommendations to improve patient care and resource allocation.

- Built cost optimization and fraud detection models integrating claims, provider, and utilization datasets to surface anomalous billing patterns and financial leakage, Built SQL models in **DBT** following a software engineering-based **CI/CD** process, designing and implementing automated tests to ensure data quality, reliability & scalability.
- Utilizing Python (Pandas, NumPy) and SQL for **EDA**, extracting valuable insights from complex Epic EHR datasets.
- Designed advanced **SQL** pipelines using **CTEs** and window functions in Snowflake to integrate claims versus **EHR** data, improving analytical consistency across reporting layers.
- Supported **CMS**-aligned reporting and analytics initiatives by validating measure logic, population definitions, and enterprise data governance controls. Implemented SQL Server Integration Services (SSIS) to optimize Extract, Transform, Load (ETL) processes. Developed automated workflows for seamless data movement into centralized repositories, enhancing data availability for analysis.
- Partnered with Data Governance teams to enforce **metadata** standards, data lineage, and quality thresholds across clinical, financial, and operational domains.

Environment: Python (Pandas, NumPy, Scikit-learn), Advanced SQL (CTEs, Subqueries, Window Functions), Snowflake, Databricks, SAS, Tableau, Power BI, Cerner, Epic Clarity, HL7/FHIR, ICD-10, CPT, HCPCS, Claims & EHR/EMR Data, Risk Prediction Models, Readmission & Fraud Detection Models, Cost Optimization Models, Descriptive Statistics, Hypothesis Testing, Correlation & Regression Analysis, Data Cleaning & Normalization, Data Governance, CMS Data Models, HIPAA

Business Data Analyst | Cognizant - INDIA

Nov 2019 – Nov 2022

- Delivered the Customer Segmentation & Market Sizing project using R and advanced **SQL**, applying data cleaning, validation, transformation, and normalization techniques to improve segment accuracy by 28% across large datasets.
- Built clustering and **predictive analytics** models in Python using Scikit-learn, supported by descriptive statistics, correlation analysis, and basic regression, strengthening personalized targeting strategies.
- Automated MIS dashboards in **Power BI** to monitor segment performance, integrating data quality controls and governance checks to support reliable executive reporting.
- Led the Credit Risk Modeling & Delinquency Forecasting project using SQL and **SAS**, performing hypothesis testing and statistical validation to improve risk scoring accuracy across lending portfolios.
- Designed Excel Power Pivot and **Power BI** reports summarizing portfolio metrics, supporting regulatory-aligned reporting and standardized data governance practices.
- Executed the Fraud Prediction & Mitigation project using Apache Spark and Python, developing **fraud detection models** and feature engineering workflows on high-volume transactional data.
- Deployed QlikView dashboards enabling real-time fraud monitoring, supported by **model validation & performance** tracking to improve operational decision-making. Managed the Churn Management & Retention Analytics project using SQL and KNIME, engineering behavioral features and building **risk prediction models** to identify high-risk customer segments.

Environment: Python, R (Tidyverse), Advanced SQL, SAS, Apache Spark, KNIME, Power BI, Tableau, QlikView, Excel Power Pivot, Predictive Analytics, Fraud Detection Models, Risk Prediction Models, Descriptive Statistics, Hypothesis Testing, Correlation & Regression Analysis, Data Transformation & Normalization, Feature Engineering, Model Validation, Data Quality Controls, Data Governance Frameworks

EDUCATION

Master of Science in Business Analytics - University of North, USA

Bachelor of Science in Mathematics, Statistics, Computer Science - Sri Venkateshwara University, India

CERTIFICATIONS

Microsoft Certified: Power BI Data Analyst Associate

AWS Certified Cloud Practitioner (CLF-C02) Cert Prep: 1 Cloud Concepts

Python for Data Analysis

Excel Essential Training (Microsoft 365)